

# WAREHOUSE MANAGEMENT FOR MOBILE STATIONS USING RFID

S. Revathi, Mansi bhardwaj, Aryan Kulshrestha, Rahul Raj  
School of Electronics Engineering, VIT, Chennai, India

[revathi.s@vit.ac.in](mailto:revathi.s@vit.ac.in), [mansi.bhardwaj2019@vitstudent.ac.in](mailto:mansi.bhardwaj2019@vitstudent.ac.in), [aryan.kulshrestha2019@vitstudent.ac.in](mailto:aryan.kulshrestha2019@vitstudent.ac.in),  
[rahul.raj2019b@vitstudent.ac.in](mailto:rahul.raj2019b@vitstudent.ac.in)

## **ABSTRACT**

RFID (Radio Frequency Identification Technology) is a computerized comprehension process which has gained traction over the past few years and is currently being considered as an effective way of improving the handling of data record. A wide range of devices and related systems have been created to meet a wide range of uses. This article describes a novel solution for managing products in warehouses and inventories for the Telecom Industry. Despite the availability of manual and automated methods for tracking and managing inventories, the use of RFID tags and scanners had not been implemented in mobile site locations. The objective of this research is to enhance the efficiency and effectiveness of the overall system. To address the practical challenges of improper warehouse management, we have developed a user-friendly graphical interface (GUI) coupled with a hardware setup capable of scanning RFID tags. This hardware system facilitates the installation of the product and its relevant information into a centralized database. The proposed solution promises to streamline the warehouse management process, reduce errors, and improve productivity. In addition to developing a user-friendly GUI and hardware setup for scanning RFID tags, our solution also includes a novel approach to secure the data collected through encryption. We recognize the critical nature of the data collected in the warehouse management system and the need to protect it from unauthorized access.

**Keywords:** RFID, Barcode, GUI, Warehouse management, Telecom industry

## **1. INTRODUCTION**

RFID technology, which is simultaneously efficient and cost effective, can be of great assistance in tracking and monitoring. Using RFID tags to uniquely identify a person or an object is achieved easily. The whole apparatus is easily scalable to achieve higher distance ranges by using higher grade tags or by using battery powered active tags[1]. A key challenge in Radio Frequency Identification is to provide complete coverage using a minimum number of readers. RFID technology is rapidly developing to track physical objects in real-time. RFID scanners and tags of various kinds are now accessible. In the Telecom Industry, efficient inventory management is crucial to ensure that the right products are available at the right site locations. However, managing inventories across multiple site locations can be a challenging task. When products are dispatched from the warehouse to site locations or defective products are returned, it becomes difficult for the telecom company to keep track of what type of product is stored at each site location. As a result, inventory clashes can occur, leading to delays in installations and increased operational costs. For instance, consider a scenario where three site locations are named A, B, and C. Suppose the company needs a particular product installed at site A, but the product is available at site B. Due to improper inventory tracking, the company may send a person to site C, located far away from both A and B, to retrieve the product manually. Such inefficiencies can lead to significant losses for the company. To address these challenges, our research proposes a solution that utilizes RFID tags and scanners to manage inventories efficiently. Compared to other inventory tracking methods like barcodes or Sapid, RFID has several advantages. RFID allows for real-time tracking of products, is faster and more accurate than barcodes, and can be used for long-distance tracking, which is not possible with Sapid. Overall, our solution aims to improve the effectiveness and performance of the entire warehouse management system by streamlining the site location inventory tracking process, thereby reducing inventory clashes, improving productivity, and minimizing operational cost.

## 2. PROPOSED METHODOLOGY

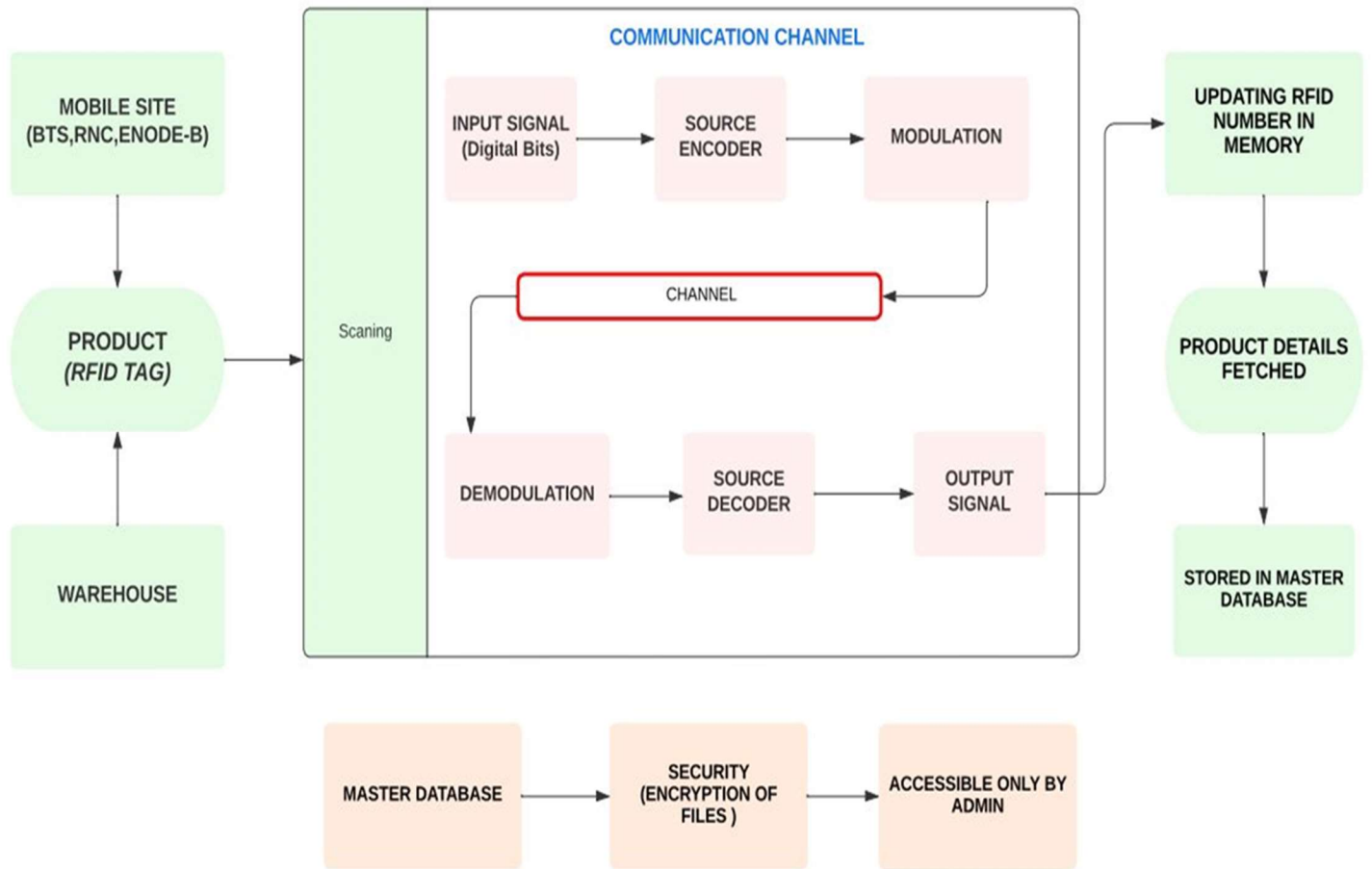


Fig.1. Project Flow Diagram

### 2.1 Hardware

- RC522: RFID READER WRITER MODULE
- Arduino Uno : 2 modules are used for scanning and uploading the data
- Circuit Connections: were done through Jumper wires and breadboard.
- RFID Tags: Predefined tags with unique ID number used for scanning purpose

The hardware used in this project consists of a reader and scanner circuit interfaced with an Arduino board. The reader is responsible for sending out an electromagnetic signal to the RFID tag and receiving the signal back from the tag. The scanner circuit is responsible for decoding the signal received from the reader and extracting the unique identifier of the RFID tag. The Arduino board acts as an intermediary between the reader/scanner circuit and the database. The board is programmed to read the data from the scanner circuit, establish a connection with the database using a JDBC driver, and update the database with the inventory information. In addition to the reader, scanner, and Arduino board, the hardware setup also includes power supplies and cables to connect the different components together. The hardware setup is designed to be compact and easy to transport, making it suitable for use in a variety of settings, including mobile site locations.

## 2.2 CIRCUIT SCHEMATIC

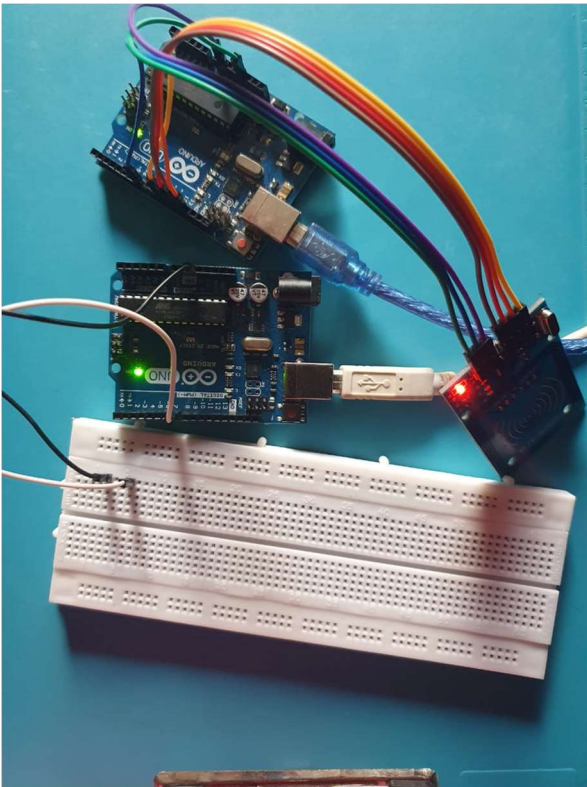


Fig.2. Circuit in Ideal situation

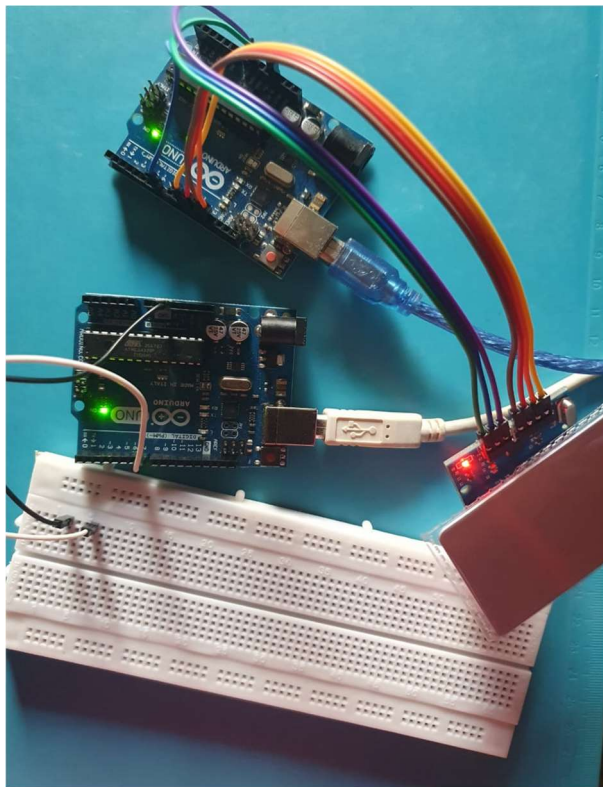


Fig.3. Circuit with RFID Tag

## 2.3 Hardware Pinout Connection

| Arduino | RC522 |
|---------|-------|
| Pin D7  | RST   |
| Pin D10 | SDA   |
| Pin D11 | MOSI  |
| Pin D12 | MISO  |
| Pin D13 | SCK   |
| GND     | GND   |
| 3.3V    | 3.3V  |

## 2.4 Software

NetBeans, PhpMyAdmin, Xampp are the software's used to integrate all hardware as well as software components of our project. We aim to develop a proper embedded system with utmost efficiency for effective and ease adaptivity for the telecom industry. The software used in this project consists of two main components: the Graphical User Interface (GUI) and the database management system. The GUI is developed using Java programming language and provides an intuitive and user-friendly interface for the users to interact with the database. The GUI allows the users to login using their unique username and password, view and manage inventory data, and generate bills for products that exit the warehouse. The database management system is developed using MySQL, which is an open-source relational database management system. The database stores all the inventory data collected from the RFID tags and is managed through a PhpMyAdmin web interface. The data is encrypted using the AES symmetric encryption algorithm to ensure security. The Java application communicates with the database using a JDBC driver, allowing the inventory data to be seamlessly integrated

into the GUI. The software architecture is designed in such a way that it can be easily modified and extended to accommodate new functionalities.

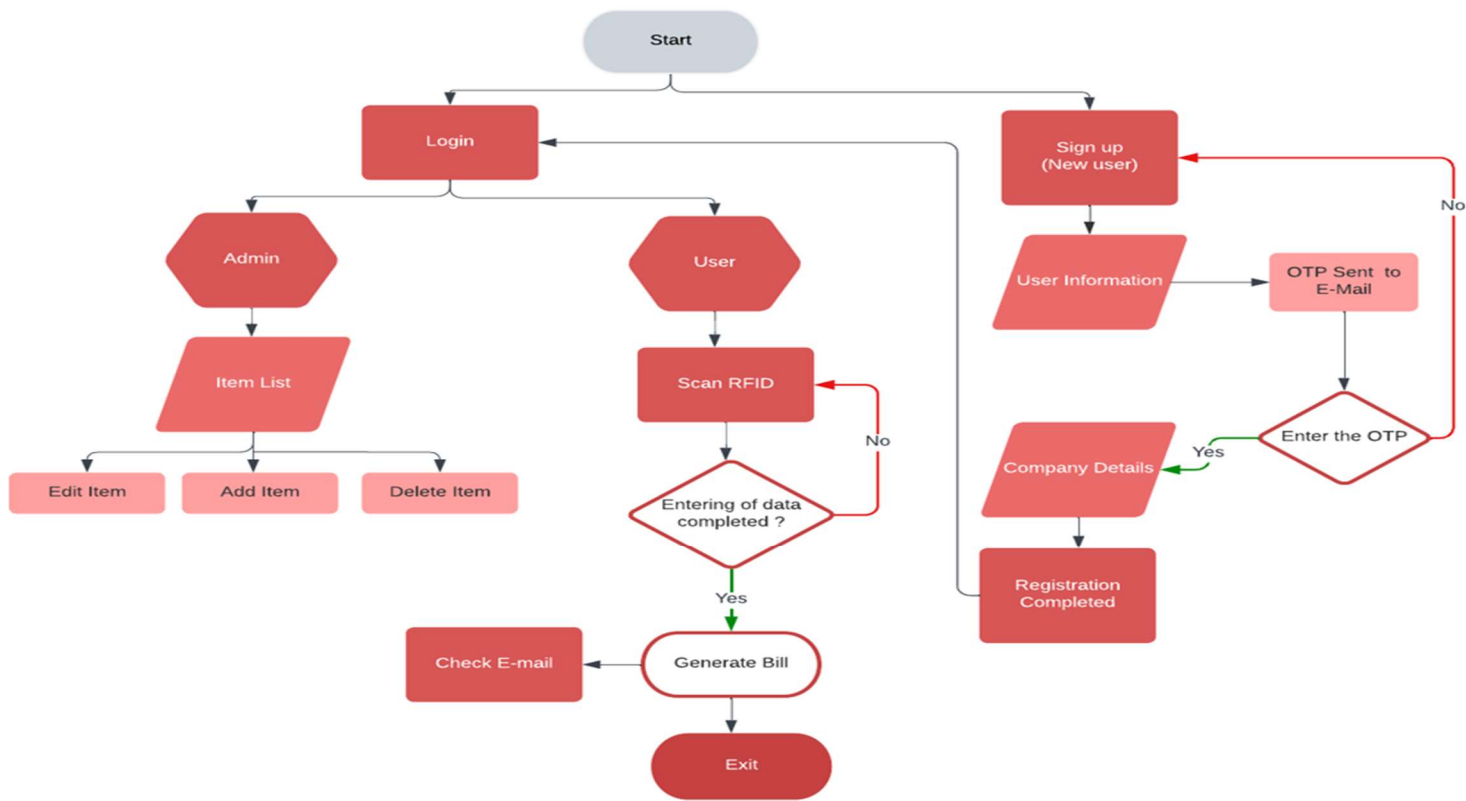


Fig.4. Work Flow of GUI

## 2.5 Orthogonal dataset

Two vectors are orthogonal if their dot product is zero. The dot product of two vectors is defined as the sum of the products of their corresponding components. So, to check if two vectors are orthogonal, compute their dot product, and if it is zero, the vectors are orthogonal. Otherwise, they are not orthogonal. To convert RFID data to a vector representation, one common approach is to use one-hot encoding. One-hot encoding is a technique used to represent categorical variables as binary vectors. In this approach, each unique RFID code is assigned a unique index or position in the vector, and the vector is set to 1 at the corresponding index and 0 elsewhere.

## 2.6 Technology and architecture

Radio Frequency Identification is a technology that uses radio waves to automatically identify and track objects. The basic components of an RFID system are an RFID tag (or transponder), an RFID reader (or interrogator), and a backend system that manages and processes the data collected by the reader. The RFID tag consists of a small microchip and an antenna that are mounted on a substrate. When the tag comes within the range of an RFID reader, the reader emits a radio signal that is received by the tag, which then responds with its unique identification code or other data stored on the chip. The reader captures this data and sends it to the backend system for processing. Now, SAP ID is a unique identification code that is assigned to each material or product in an SAP system. It is used to track and manage inventory, and to facilitate the movement of goods through the supply chain. SAP ID can be linked to an RFID tag or a barcode for easy identification and tracking. On the other hand, a barcode is a visual representation of data that is typically printed on a label or tag. It consists of a series of

parallel bars and spaces of varying widths that encode information such as a product code, serial number, or other data. Barcodes are read by scanning them with a barcode scanner, which translates the visual pattern into digital data that can be processed by a computer system. Compared to barcodes, RFID tags have some advantages such as they do not require line-of-sight scanning, can be read from a distance, and can be used in harsh environments. RFID tags also have a higher storage capacity than barcodes and can store more data, such as the item's manufacturing date, expiration date, and location history. In summary, while both RFID and barcodes are used for identification and tracking, RFID offers some advantages over barcodes in terms of functionality, especially in situations where the objects being tracked are located in difficult-to-reach places or need to be read from a distance. SAP ID can be used with both RFID tags and barcodes to facilitate the tracking and management of inventory and other assets in an SAP system.

## **2.7 Security features for our proposed model**

### **2.7.1 AES Encryption Decryption**

In this project, the inventory data stored in the MySQL database is encrypted using the Advanced Encryption Standard (AES) symmetric encryption algorithm. This is done to ensure the security and confidentiality of the data stored in the database. AES is a widely used encryption algorithm that uses a symmetric key to encrypt and decrypt data. In symmetric encryption, the same key is used for both encryption and decryption, and therefore, the key must be kept secret to ensure the security of the data. In this project, the encryption and decryption of the data is done by the Java application using the Java Cryptography Extension (JCE) library. The JCE library provides a set of classes and interfaces for performing encryption and decryption operations. When the inventory data is inserted into the database, it is encrypted using a randomly generated secret key. The secret key is then encrypted using the RSA algorithm and stored in a separate table in the database. This ensures that the secret key used for encryption and decryption is kept secure and is not visible to any unauthorized user. When the inventory data is retrieved from the database, the secret key is first decrypted using the RSA algorithm and then used to decrypt the encrypted inventory data. This ensures that the inventory data can only be accessed by authorized users who have access to the secret key. Overall, the encryption and decryption of the inventory data ensures the confidentiality and security of the data stored in the database, and helps to protect against unauthorized access or data breaches.

### **2.7.2 OTP and Email**

When a new user wants to sign up for the system, they need to enter their email address and a password. After submitting the form, the system generates a unique OTP and sends it to the user's email address. The user is then required to enter the OTP in the application to complete the sign-up process. The OTP is a random, time-sensitive code that is only valid for a short period of time. It is generated using an algorithm that ensures that the code is unique and cannot be reused. This ensures that only the user who has access to the email address can complete the sign-up process. The OTP sent on email adds an additional layer of security to the application, as it ensures that only authorized users can access the system. It helps to protect against unauthorized access and data breaches. Overall, the OTP feature is an important part of the system's security architecture, and helps to ensure that the data stored in the system is protected against unauthorized access.

### 2.7.3 Digital Signature with MD5 Hash code

The purpose of including a digital signature in a bill is to ensure the authenticity and integrity of the document, as any changes to the document will result in a different signature. To generate a digital fingerprint for a bill, we have used the MD5 function. When verifying the authenticity of the bill, the digital fingerprint can be recalculated from the bill and compared to the original digital fingerprint to ensure that the bill has not been tampered.

## 3. GRAPHICAL USER INTERFACE

The content explains the development of a software system that uses Java programming language to create a Graphical User Interface (GUI) for users to interact with a database. The GUI allows users to login using unique credentials and perform various tasks such as viewing and managing inventory data, generating bills for products that leave the warehouse, and other related functions. The database management system is created using MySQL, which is an open-source relational database management system. This system stores all the inventory data collected from the RFID tags and is managed through a PhpMyAdmin web interface. The data is encrypted using the AES symmetric encryption algorithm to ensure that it remains secure.

To ensure seamless integration of inventory data into the GUI, the Java application communicates with the database using a JDBC driver. This allows for the easy integration and manipulation of data between the GUI and the database. The software architecture is designed in a way that it can be easily modified and extended to accommodate new functionalities.

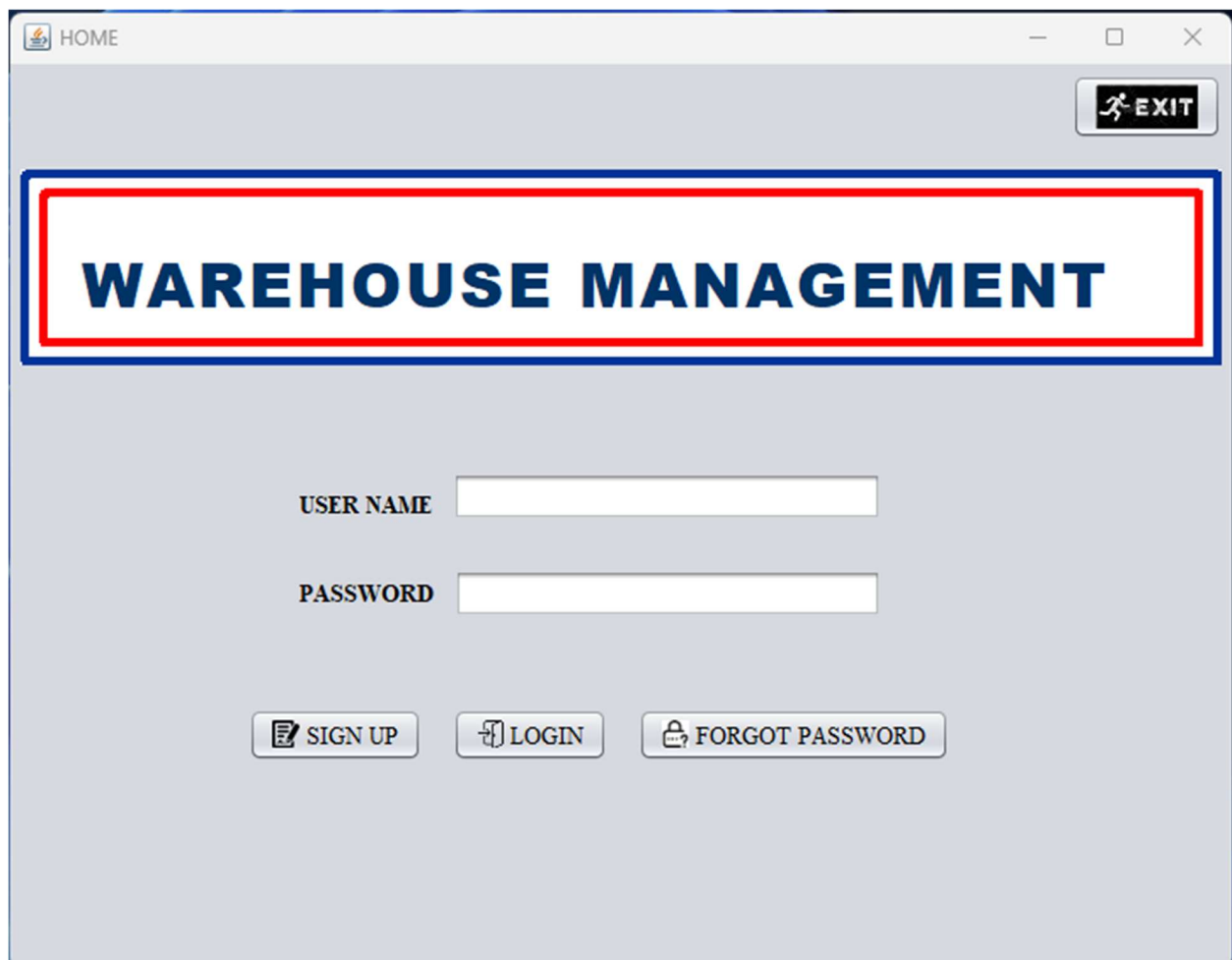


Fig.5. Home page



REGISTRATION

BACK EXIT

# WAREHOUSE MANAGEMENT

## USER INFORMATION

NAME

GENDER ☐ MALE ☐ FEMALE

E-MAIL

ADDRESS

CITY

DATE OF BIRTH

☐ I AGREE WITH ALL YOUR TERMS AND CONDITION

☐ HERE BY I DECLARE ALL THE ABOVE INFORMATION TO BE TRUE

>> NEXT

Fig.6. Signup

REGISTRATION

BACK EXIT

# WAREHOUSE MANAGEMENT

## USER INFORMATION

NAME

GENDER ☐ MALE ☐ FEMALE

E-MAIL

ADDRESS

CITY

DATE OF BIRTH

☒ I AGREE WITH ALL YOUR TERMS AND CONDITION

☒ HERE BY I DECLARE ALL THE ABOVE INFORMATION TO BE TRUE

>> NEXT

Message

\*\*\* OTP Send To Your Registered Email Successfully \*\*\*

OK

Fig.7. One time password is sent to the Email

ADD ITEM

BACK

EXIT

WAREHOUSE MANAGEMENT

RFID NO.

ITEM

LOCATION

0

ADD ITEM

| RFID NO.       | ITEM        | LOCATION    |
|----------------|-------------|-------------|
| 49 ED 4F 73    | ROUTER      | SUBSTATION1 |
| 6E 6B 45 73    | Fiber cable | SUBSTATION2 |
| 41 F1 97 B9    | Antenna     | substation3 |
| /4 @? /A 2,    | hj_`h       | //          |
| /4 @M /A 2,    | Vo`il       | ..*         |
| /4 B? /A 2,    | adj`m       | ~J          |
| /4 @H /A 2,    | `\jg`       | ~J          |
| 0,-j`+0/3/2//, | adj`m       | 3           |

Fig.8. ADD Button - To add new details to the product list

BACK

EXIT

WAREHOUSE MANAGEMENT

RFID NO.

ITEM

LOCATION:

UNIT :

EDIT

APPLY

| RFID NO.       | ITEM        | LOCATION    |
|----------------|-------------|-------------|
| 49 ED 4F 73    | ROUTER      | SUBSTATION1 |
| 6E 6B 45 73    | Fiber cable | SUBSTATION2 |
| 41 F1 97 B9    | Antenna     | substation3 |
| /4 @? /A 2,    | hj_`h       | //          |
| /4 @M /A 2,    | Vo`il       | ..*         |
| /4 B? /A 2,    | adj`m       | ~J          |
| /4 @H /A 2,    | `\jg`       | ~J          |
| 0,-j`+0/3/2//, | adj`m       | 3           |

Fig.9. EDIT Button - To edit the details of the existing product list



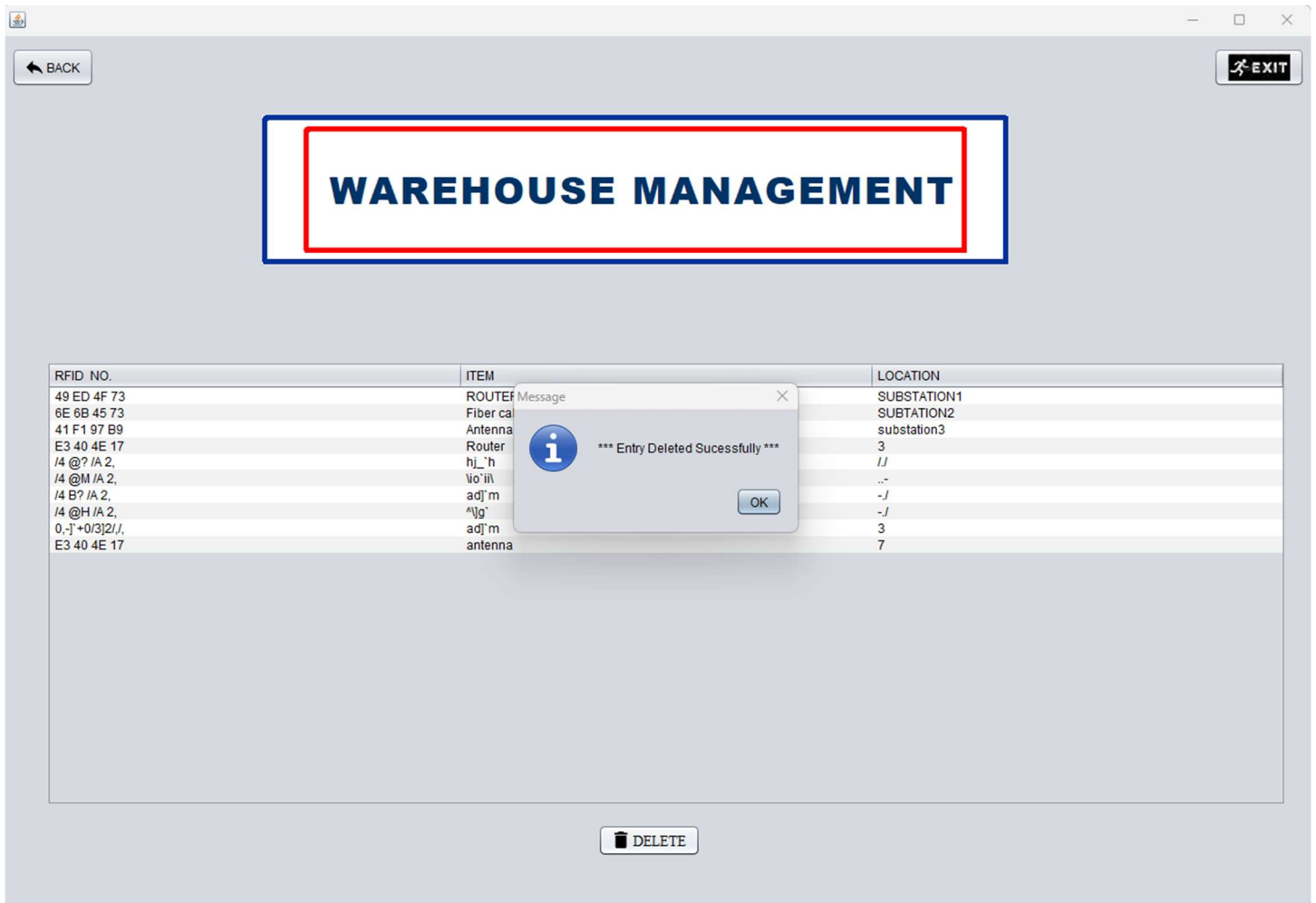


Fig.10.DELETE Button - To delete an entry from the product list

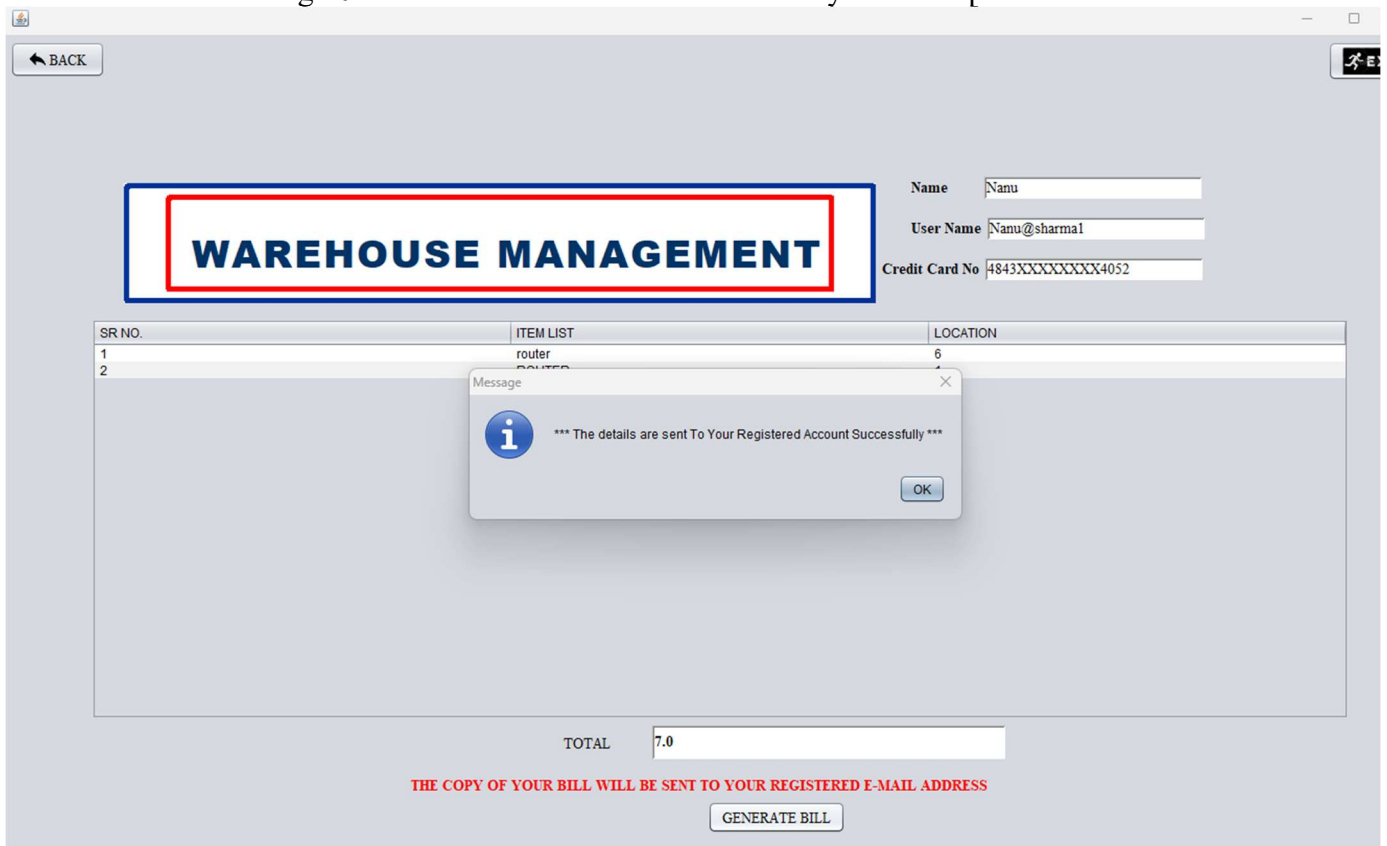


Fig.11.Bill has been sent to Email

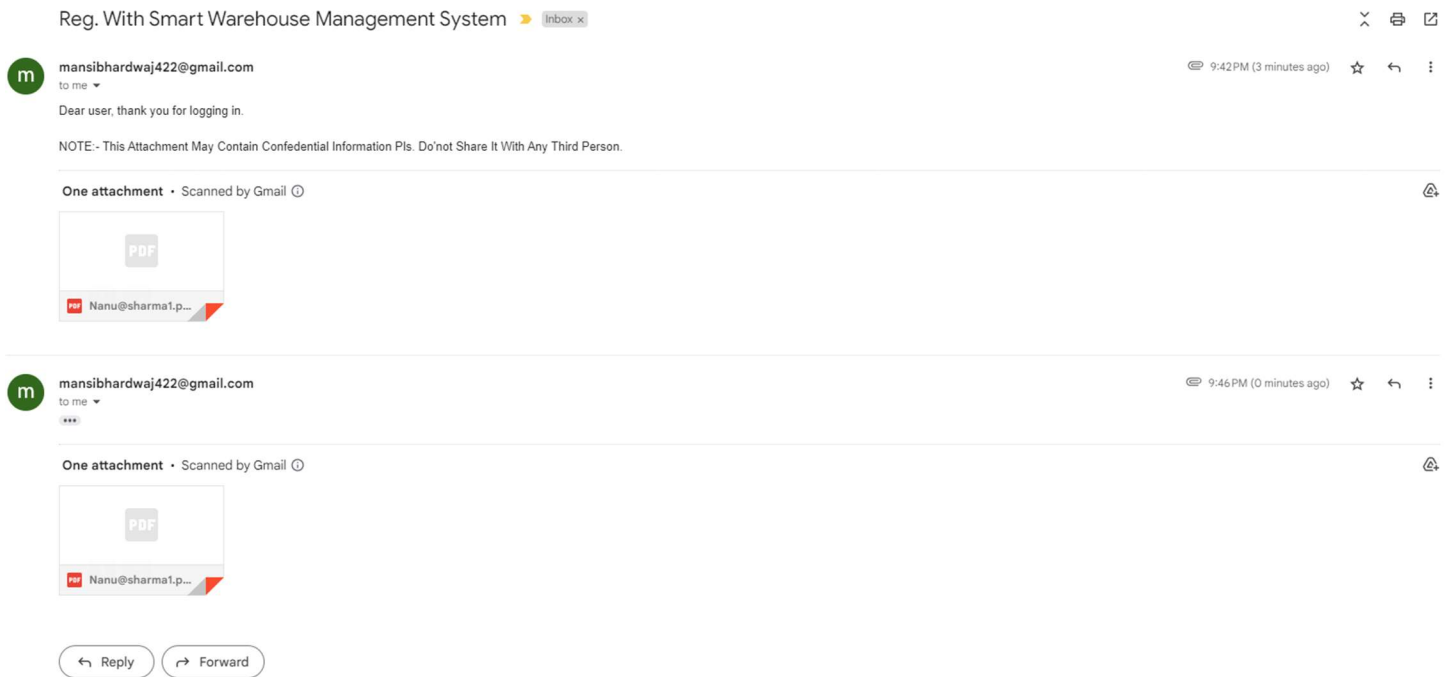


Fig.12.Bill Received on Registered Email

The overall content highlights the development of a software system that uses RFID technology to collect inventory data and a GUI created using Java programming language to interact with the database management system.

#### 4. RESULTS

The results of our research demonstrate the potent of utilizing RFID tags and scanners to manage inventories in the Telecom Industry. The concept of managing products via the warehouse led us to focus on scanning RFID tags, which provide a unique identifier for each item in the warehouse. This approach allows for precise tracking of products, thereby reducing inventory clashes and improving the overall efficiency of the inventory management system. To ensure the privacy and security of the data collected, we designed a user-friendly graphical interface using Java, which is connected to an SQL database hosted on a PhpMyAdmin server. Access to this database is restricted to only authorized personnel, and new users must go through an OTP/email verification process before being granted access. The GUI enables users to log in using a username and password and has functions to add, delete, and edit data using RFID tags. Additionally, when products exit the warehouse, a bill is generated and sent to the user's email, ensuring that all transactions are recorded accurately. To validate the real-time functionality of our solution, we developed a hardware system that includes reader and scanner circuits interfaced with an Arduino. The Arduino is connected to a JDBC database, which sends data to a MySQL server. The MySQL database is encrypted using AES symmetric encryption to ensure that sensitive information is protected from unauthorized access. This hardware system accurately senses RFID tags within the line of sight and updates the database in real-time, ensuring that inventory information is always up-to-date.

#### 5. CONCLUSION AND FUTURE WORK

In conclusion, our research demonstrates the effectiveness of utilizing RFID tags and scanners to manage inventories in the Telecom Industry. Our solution provides a secure, efficient, and accurate approach to inventory management that can reduce operational costs and improve productivity. The combination of a user-friendly GUI and a real-time hardware system has the potential to revolutionize inventory management in the Telecom Industry and beyond. The current system focuses on inventory management at mobile site locations using RFID. In the future, the system could be integrated with other systems, such as supply chain management or logistics, to provide a more comprehensive solution. Machine learning algorithms could be integrated into the system to

predict inventory levels and demand based on historical data. This could help optimize inventory levels and reduce waste.

## 6. REFERENCES

- [1] A. Agarwal, G. Babu, L. Rishi, K. Shruthi and V. Agarwal, "RFID management for tracking and inventory," 2011 Second Asian Himalayas International Conference on Internet (AH-ICI), Kathmandu, Nepal, 2011, pp. 1-3, doi: 10.1109/AHICI.2011.6113935.
- [2] Ghelichi, A., & Abdelgawad, A. (2014, December). A study on RFID-based Kanban system in inventory management. In 2014 IEEE International Conference on Industrial Engineering and Engineering Management (pp. 1357-1361). IEEE.
- [3] Al Kattan, I., & Al-Khudairi, T. (2007, July). Improving supply chain management effectiveness using RFID. In 2007 IEEE International Engineering Management Conference (pp. 191-198). IEEE.
- [4] Agarwal, A., Babu, G., Rishi, L., Shruthi, K. S., & Agarwal, V. (2011, November). RFID management for tracking and inventory. In 2011 Second Asian Himalayas International Conference on Internet (AH-ICI) (pp. 1-3). IEEE.
- [5] Jhunjhunwala, P., Shriya, M., & Rufus, E. (2019, March). Development of hardware based inventory management system using uav and rfid. In 2019 International Conference on Vision Towards Emerging Trends in Communication and Networking (ViTECoN) (pp. 1-5). IEEE.
- [6] Cristiani, D., Bottonelli, F., Trotta, A., & Di Felice, M. (2020, August). Inventory management through mini-drones: Architecture and proof-of-concept implementation. In 2020 IEEE 21st International Symposium on "A World of Wireless, Mobile and Multimedia Networks"(WoWMoM) (pp. 317-322). IEEE.
- [7] Bose, R., Mondal, H., Sarkar, I., & Roy, S. (2022). Design of smart inventory management system for construction sector based on IoT and cloud computing. *e-Prime-Advances in Electrical Engineering, Electronics and Energy*, 2, 100051.
- [8] Saksena, A. K., & Agarwal, R. (2021, January). Methods for Classification of Items for Inventory Management. In 2021 International Conference on Computer Communication and Informatics (ICCCI) (pp. 1-4). IEEE.
- [9] Roodbergen, K. J., & Vis, I. F. (2009). A survey of literature on automated storage and retrieval systems. *European journal of operational research*, 194(2), 343-362.
- [10] Teja, P. R., & Kumaar, A. N. (2018, September). QR code-based path planning for warehouse management robot. In 2018 International Conference on Advances in Computing, Communications and Informatics (ICACCI) (pp. 1239-1244). IEEE.
- [11] Agarwal, A., Babu, G., Rishi, L., Shruthi, K. S., & Agarwal, V. (2011, November). RFID management for tracking and inventory. In 2011 Second Asian Himalayas International Conference on Internet (AH-ICI) (pp. 1-3). IEEE.
- [12] Williams, B. D., & Tokar, T. (2008). A review of inventory management research in major logistics journals: Themes and future directions. *The International Journal of Logistics Management*, 19(2), 212-232.